

AlphaZero & MuZero

- AlphaZero, a generalized version of AlphaGo Zero
Took 4 hours to learn Chess and defeat reigning world computer chess champion 28 to 0 in 100 matches
- https://www.youtube.com/watch?time_continue=7&v=tXIM99xPQC8
- MuZero: Master games without knowing rules
- Uses approach similar to AlphaZero, developed in 2019
- Trained via self-play and play against AlphaZero with no access to rules, opening books or endgame tables
- Viewed as significant advancement over AlphaZero

AlphaFold: solution to a 50 year old grand challenge in biology

- <https://deepmind.com/blog/article/alphafold-a-solution-to-a-50-year-old-grand-challenge-in-biology>
- Figuring out what shapes proteins fold into is known as the “protein folding problem” - grand challenge in biology for the past 50 years
- Focus of intensive scientific research for many years, using a variety of experimental techniques such as nuclear magnetic resonance and X-ray crystallography.

AlphaFold

- Number of ways a protein could theoretically fold before settling into its final 3D structure is astronomical.
- Cyrus Levinthal estimated 10^{300} possible conformations for a typical protein.
- Estimated would take longer than the age of universe to enumerate all possible configurations. Yet in nature, proteins fold spontaneously, some within milliseconds - referred to as Levinthal's paradox.

Nobel Prize in Chemistry 2024 !!!

- The Nobel Prize in Chemistry 2024 is about proteins, life's ingenious chemical tools.
- David Baker has succeeded with the almost impossible feat of building entirely new kinds of proteins.
- Demis Hassabis and John Jumper have developed an **AI model** to solve a 50-year-old problem: predicting proteins' complex structures. These discoveries hold enormous potential.

Nobel Prize in Physics 2024 !!!

- They used physics to find patterns in information
- John J Hopfield and Geoffrey Hinton
- “for foundational discoveries and inventions that enable machine learning with artificial neural networks”

Turing Award in 2024 !!!

- Nobel prize of computing
- Awarded to Andrew Barto and Richard Sutton
- “for developing the foundational concepts and algorithms for **Reinforcement Learning (RL)**, a key area in Artificial Intelligence, particularly for training large models like LLMs”

AI Index 2025

- Source: Stanford Institute for Human-Centered Artificial Intelligence (HAI)
- Report: 2025 AI Index Report
- <https://hai.stanford.edu/ai-index/2025-ai-index-report>
- Slides follow

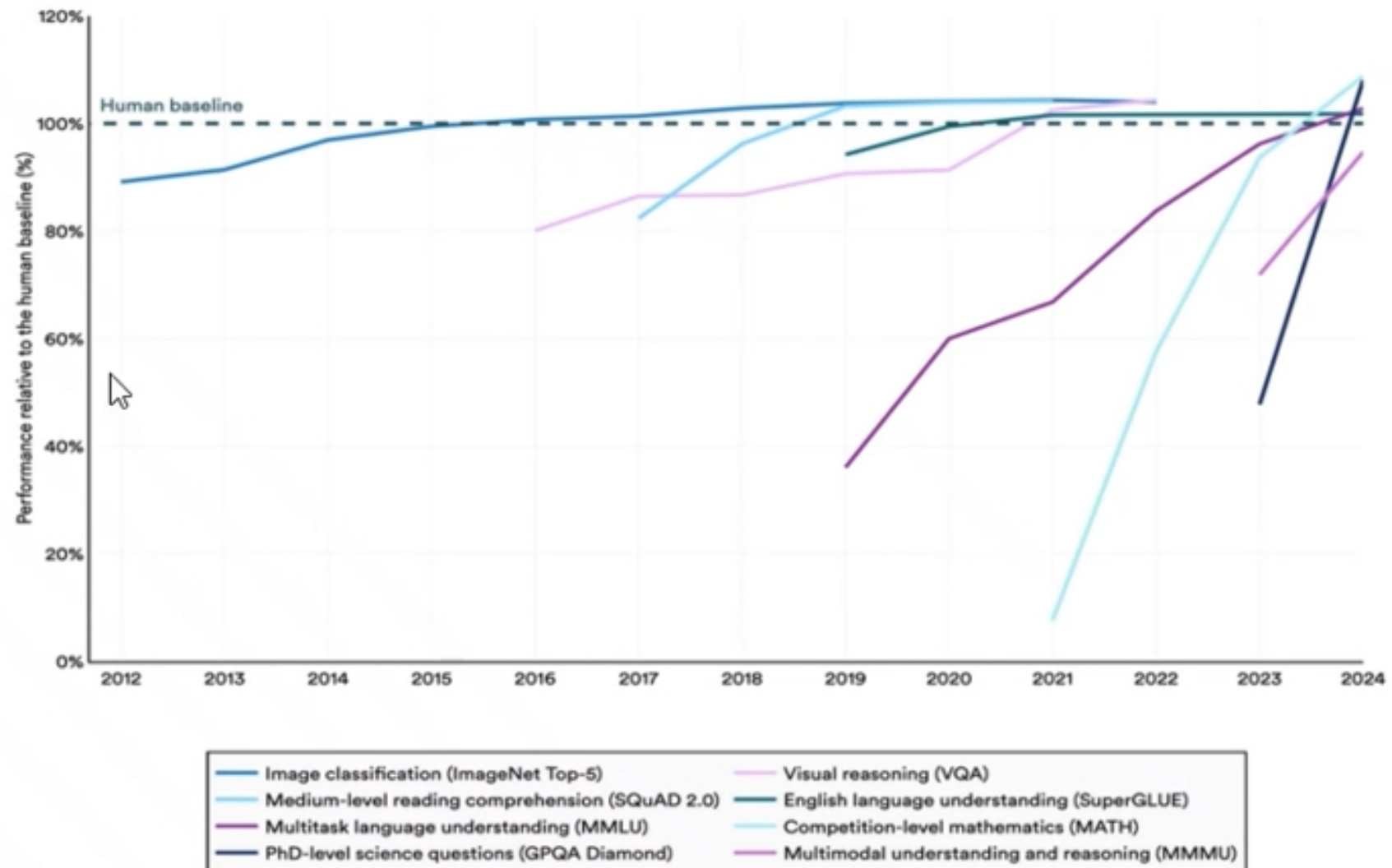
AI Performance on Demanding Benchmarks

- AI achieved large gains on harder benchmarks
- Strong progress in coding and video generation
- AI agents outperform humans in time-constrained tasks

AI Performance on Demanding Benchmarks — Data View

Select AI Index technical performance benchmarks vs. human performance

Source: AI Index, 2025 | Chart: 2025 AI Index report



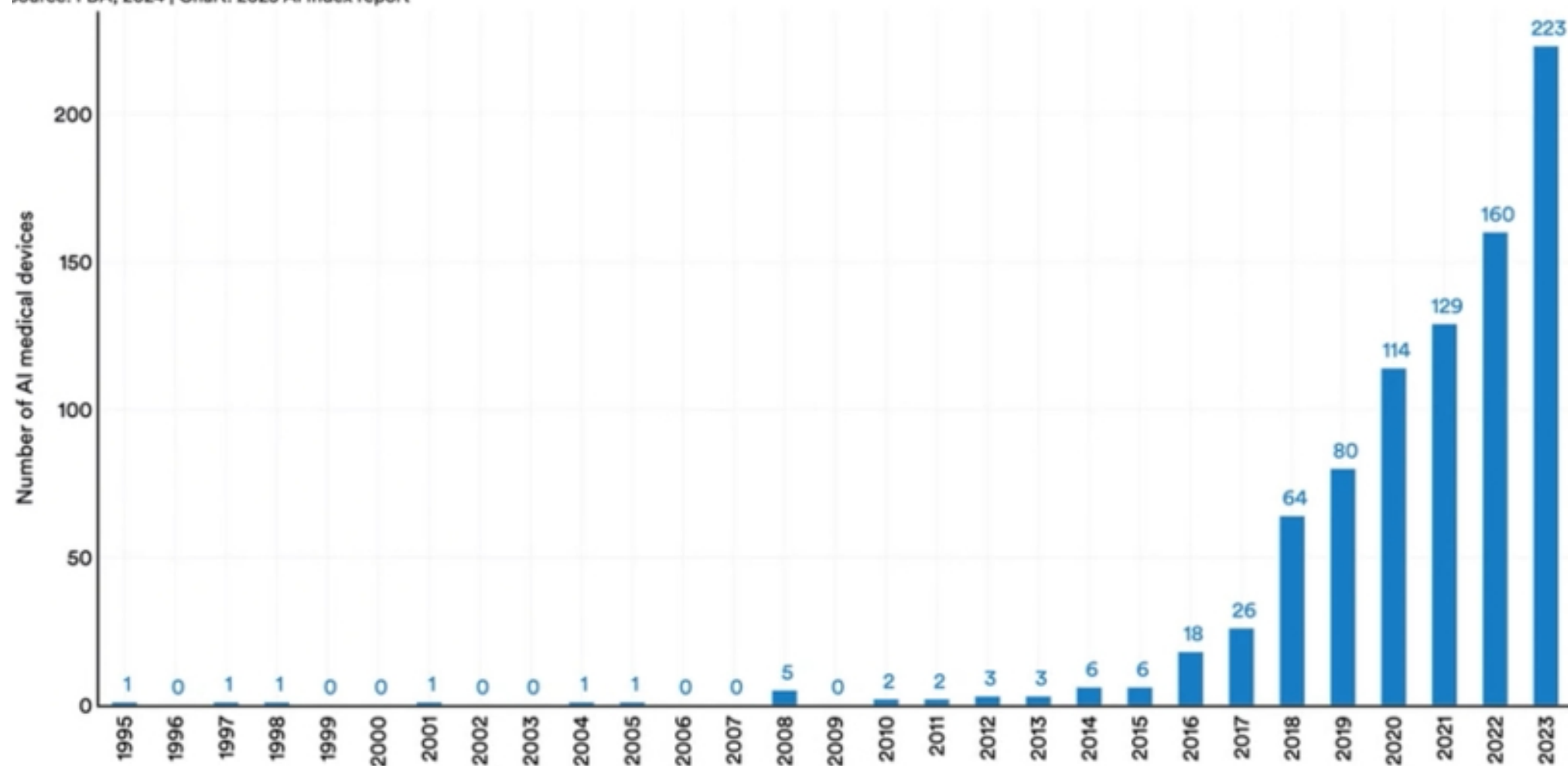
AI Embedded in Everyday Life

- AI adoption expanded across healthcare and transport
- FDA approvals of AI medical devices surged
- Autonomous vehicles reached commercial scale

AI Embedded in Everyday Life — Data

Number of AI medical devices approved by the FDA, 1995–2023

Source: FDA, 2024 | Chart: 2025 AI Index report



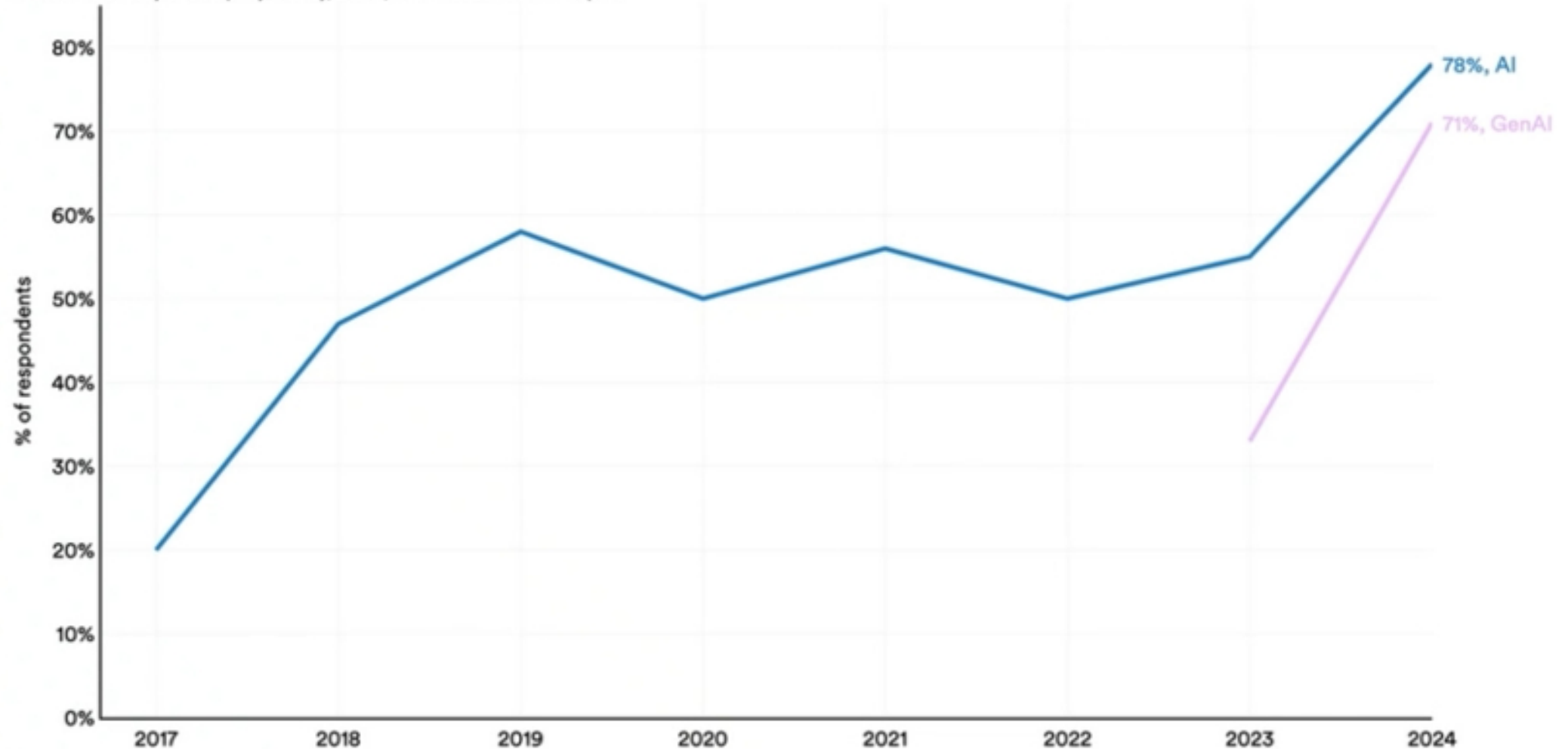
Business Investment in AI Hit New Highs

- U.S. private AI investment reached \$109.1B
- Generative AI funding remained strong
- 78% of organizations report AI use

Business Investment in AI Hit New Highs — Data View

Share of respondents who say their organization uses AI in at least one function, 2017–24

Source: McKinsey & Company Survey, 2024 | Chart: 2025 AI Index report



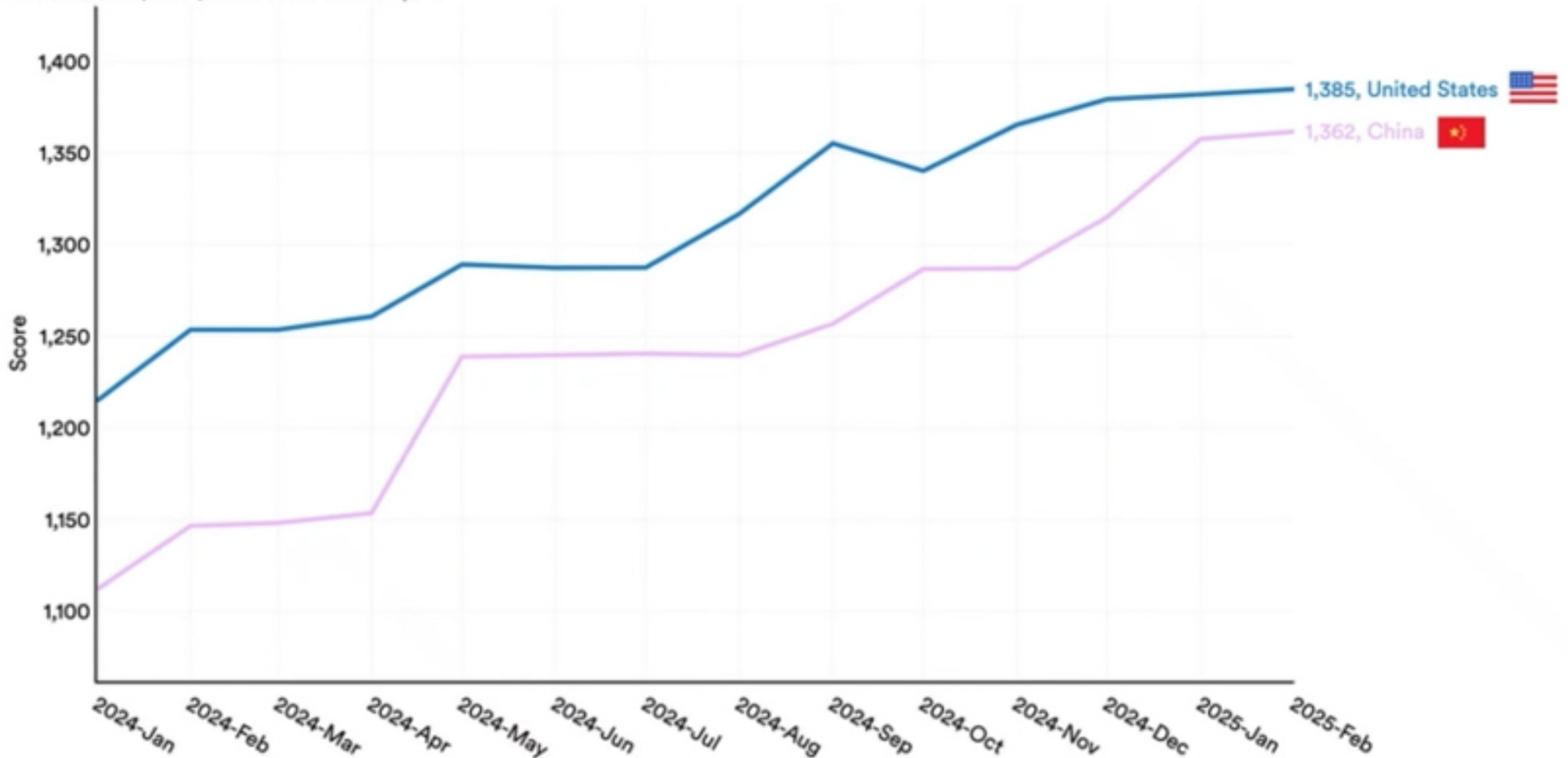
U.S. Leads, China Is Closing the Gap

- U.S. produced most notable AI models
- China narrowed performance gaps
- AI development increasingly global

U.S. Leads, China Is Closing the Gap — Data View

Performance of top United States vs. Chinese models on LMSYS Chatbot Arena

Source: LMSYS, 2025 | Chart: 2025 AI Index report



Responsible AI Progress Is Uneven

- AI incidents increased faster than safeguards
- New safety benchmarks emerged
- Governance frameworks expanded

Responsible AI Progress Is Uneven — Data View

Reported safety and responsible AI benchmarks for popular foundation models

Source: AI Index, 2025 | Table: 2025 AI Index report

Responsible AI benchmark	o1	GPT-4.5	DeepSeek-R1	Gemini 2.5	Grok-2	Claude 3.7 Sonnet	Llama 3.3
BBQ	✓	✓				✓	
HarmBench							
Cybench						✓	
SimpleQA			✓	✓			
Toxic WildChat	✓	✓				✓	
StrongREJECT	✓	✓					
WMDP benchmark	✓	✓					
MakeMePay	✓	✓					
MakeMeSay	✓	✓					

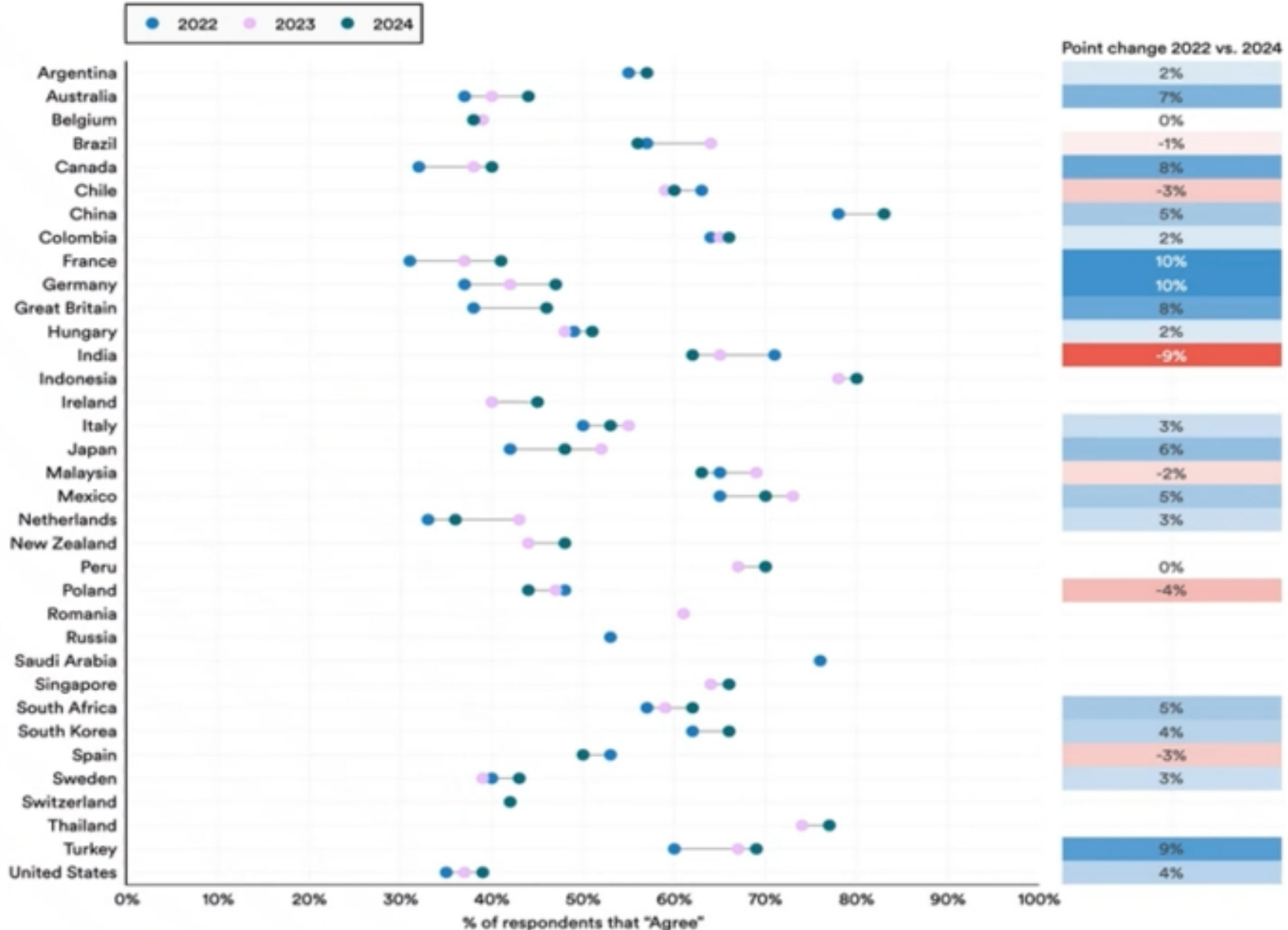
Global AI Optimism Is Rising

- Optimism strongest in Asia
- Sentiment improving globally
- Large regional differences remain

Global AI Optimism Is Rising — Data View

'Products and services using AI have more benefits than drawbacks,' by country (% of total), 2022–24

Source: Ipsos, 2022–24 | Chart: 2025 AI Index report



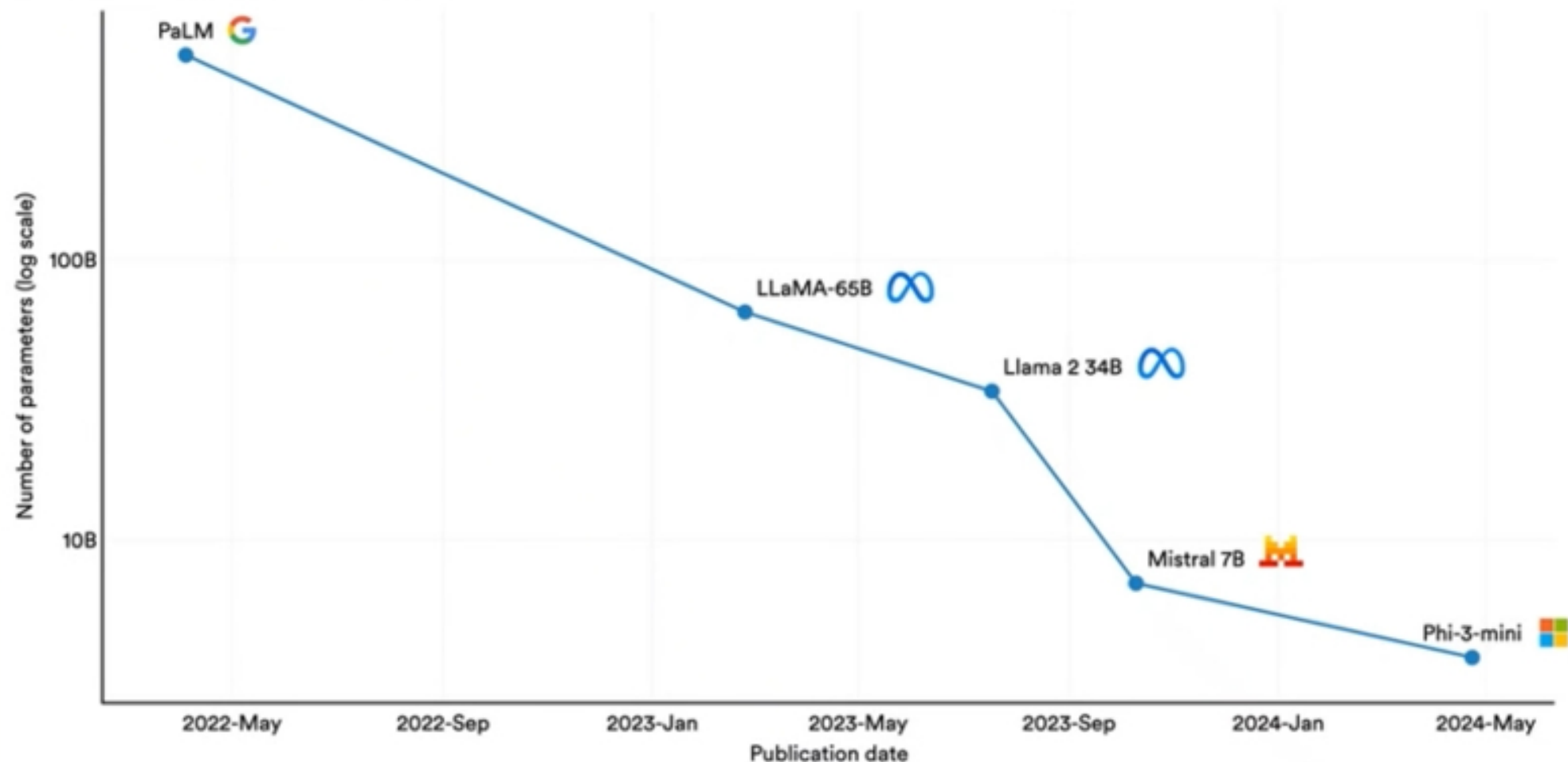
AI Became Cheaper and More Accessible

- Inference costs dropped up to 280x
- Energy efficiency improved
- Open models closed performance gaps

AI Became Cheaper and More Accessible — Data View

Smallest AI models scoring above 60% on MMLU, 2022–24

Source: Abdin et al., 2024 | Chart: 2025 AI Index report



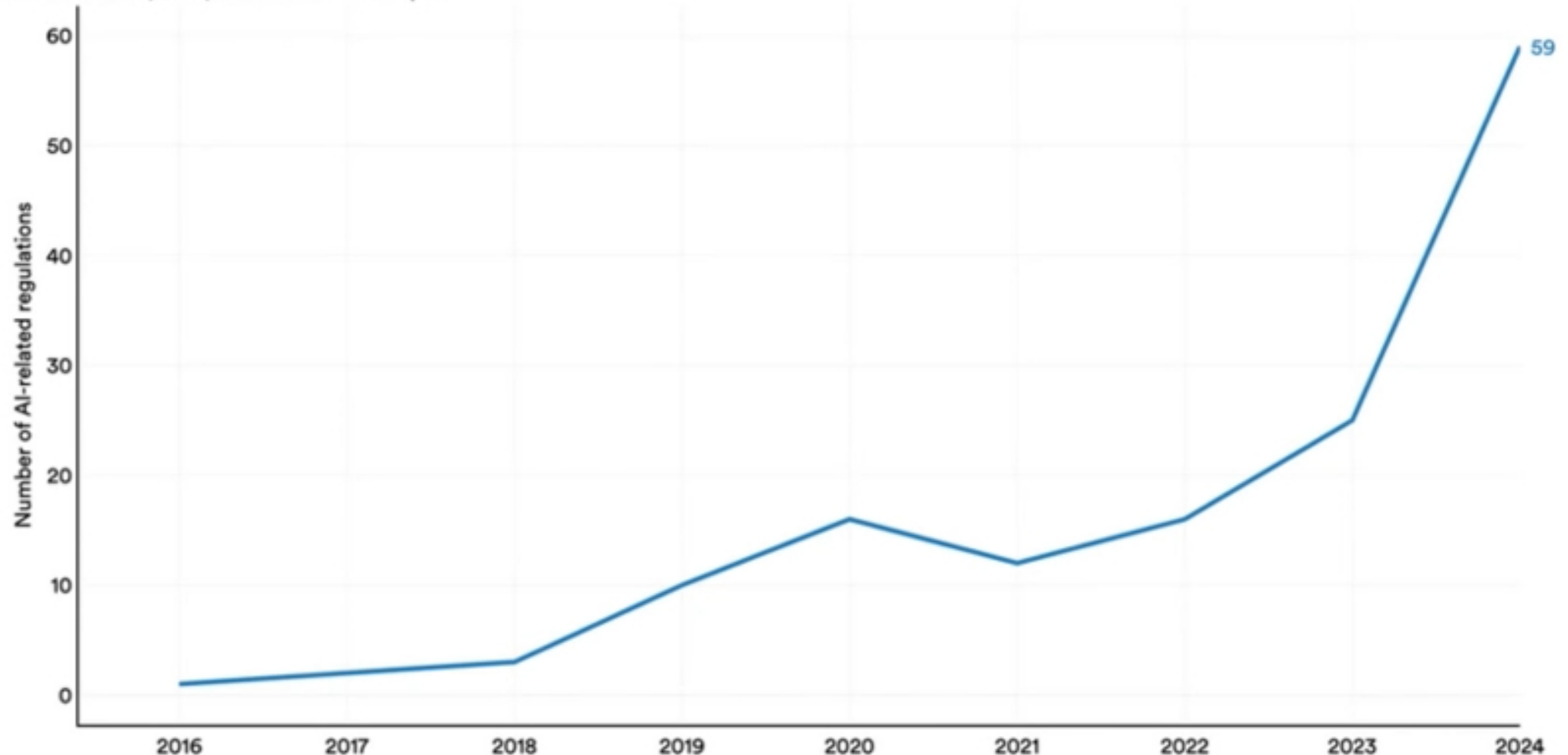
Governments Expanded AI Policy

- 59 U.S. federal AI regulations in 2024
- Global AI legislation mentions rose
- Major public investment in AI infrastructure

Governments Expanded AI Policy — Data View

Number of AI-related regulations in the United States, 2016–24

Source: AI Index, 2025 | Chart: 2025 AI Index report



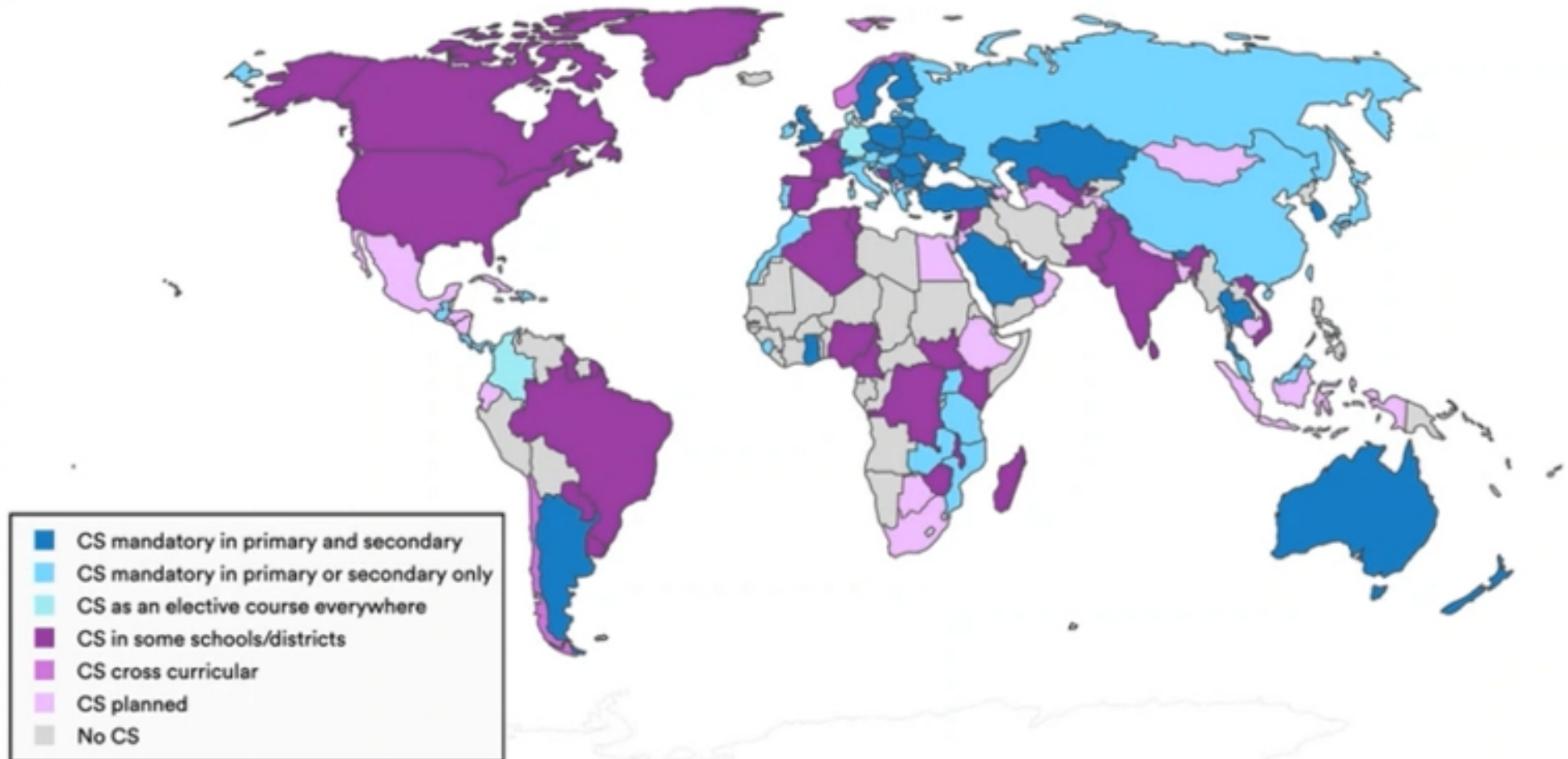
AI & CS Education Expanded

- Two-thirds of countries offer AI/CS education
- CS graduates increased in the U.S.
- Infrastructure gaps persist

AI & CS Education Expanded — Data View

Availability of CS education by country, 2024

Source: Raspberry Pi Computing Education Research Centre, 2024 | Chart: 2025 AI Index report



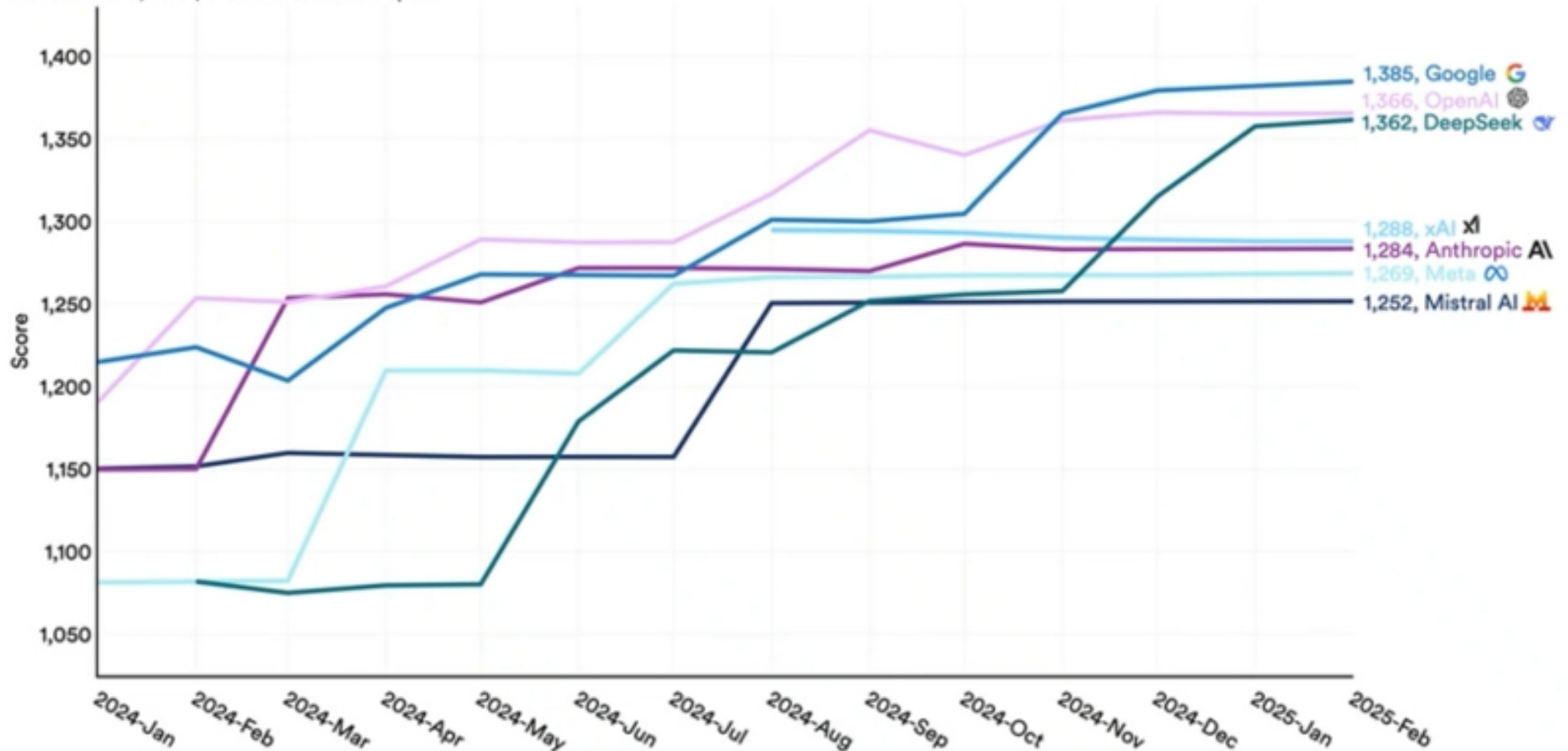
Industry Dominates AI Development

- ~90% of notable models built by industry
- Performance gaps narrowed
- Compute scale continues to rise - training compute doubles every five months, datasets every eight, and power use annually.

Industry Dominates AI Development — Data View

Performance of top models on LMSYS Chatbot Arena by select providers

Source: LMSYS, 2025 | Chart: 2025 AI Index report



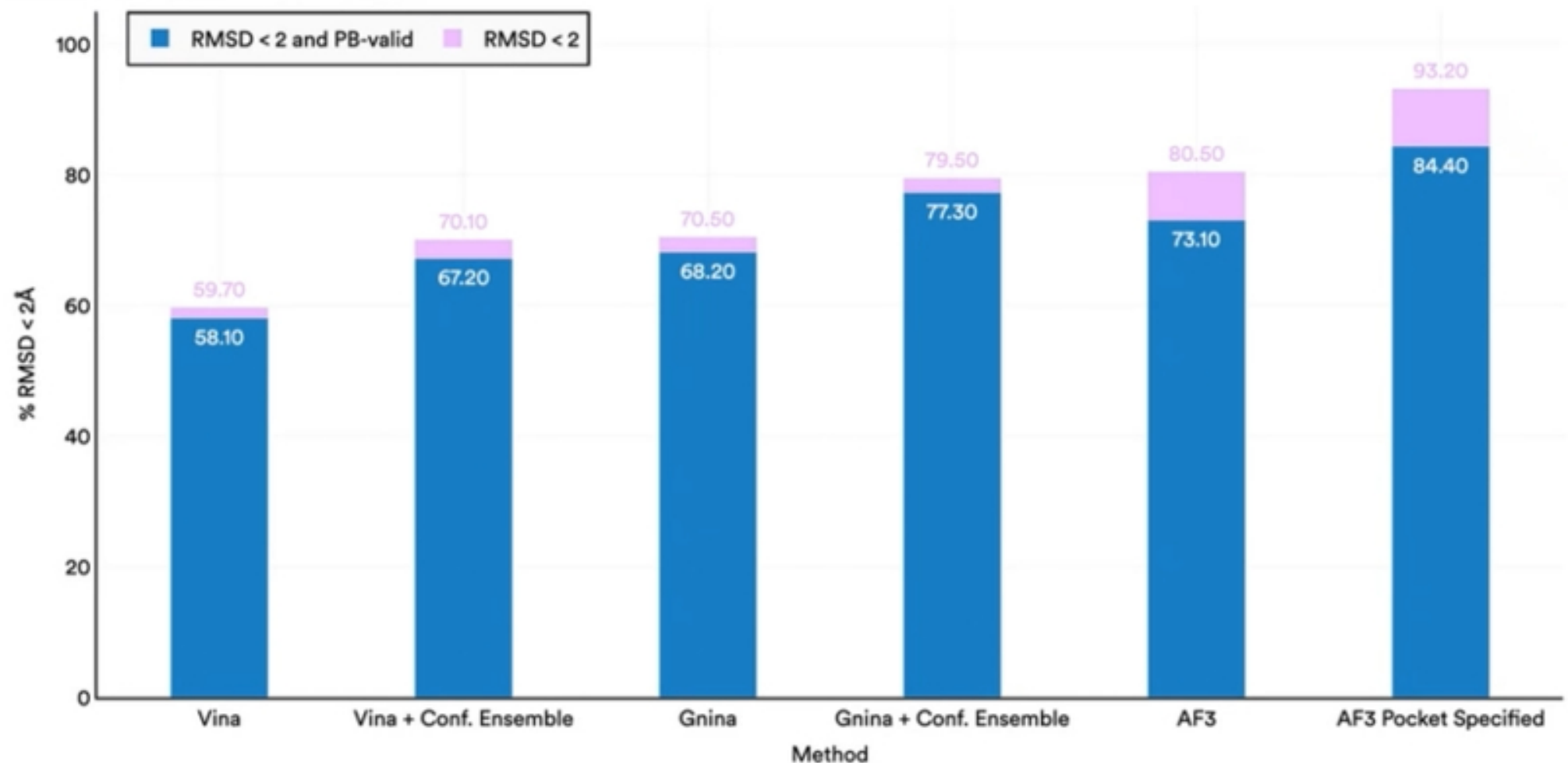
AI Earned Top Scientific Honors

- Nobel Prizes recognized AI breakthroughs
- Turing Awards highlighted reinforcement learning
- AI reshaping scientific research

AI Earned Top Scientific Honors — Data View

AlphaFold 3 vs. baselines for protein-ligand docking

Source: ESM3, 2024 | Chart: 2025 AI Index report



Complex Reasoning Remains a Challenge

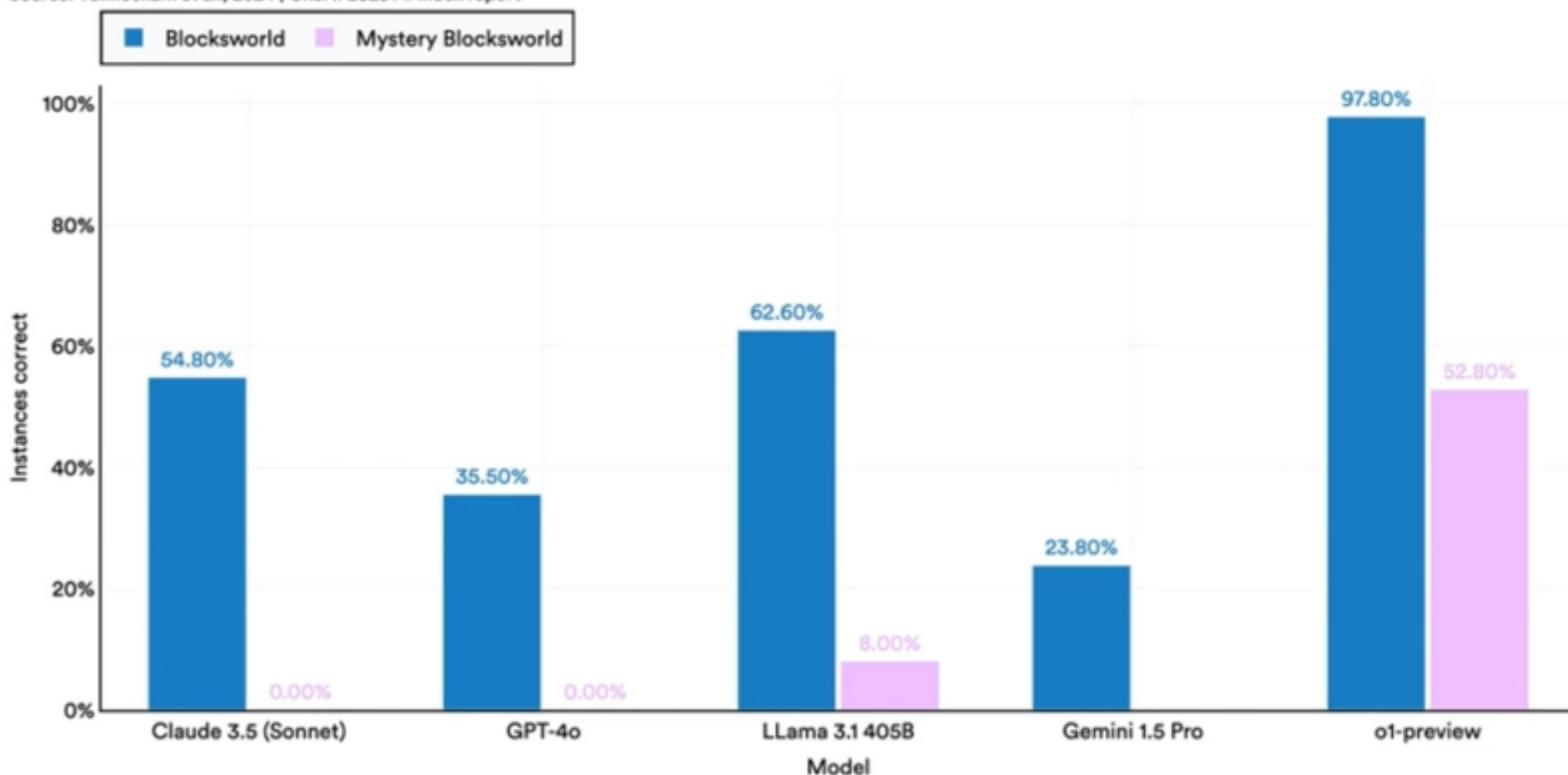
- Strong structured task performance
- Weaknesses in deep reasoning
- Limits high-stakes deployment

Complex Reasoning Remains a Challenge

— Data View

PlanBench: instances correct

Source: Valmeekam et al., 2024 | Chart: 2025 AI Index report



Overall

- Lot of developments expected going forward...